ECON 0100 | Vignette E1 Long | Costs

Solutions

Answers and work are shown in the red blocks.

As an economics student at Hogwarts, the Ministry of Magic has tasked you with analyzing the impact of policy proposals on three firms in the cauldron industry, given their distinct cost structures:

The Cauldron Co. is a large, established firm producing high-quality cauldrons with traditional methods. They have high fixed costs but low, constant marginal costs due to economies of scale.

The Witching Hour is a smaller firm specializing in novelty cauldrons. They have moderate fixed costs but increasing marginal costs due to specialized production.

Nimbus Cauldron Works is a new entrant producing low-cost, mass-produced cauldrons made to order by a contract foundry. They have no meaningful fixed costs, and low constant marginal costs.

Q1 | Cost Curves

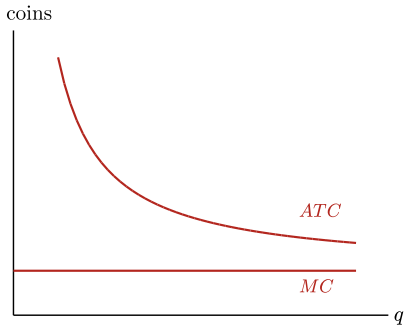
Graph all three firms' average total cost and marginal cost curves.

The Cauldron Co. Marginal cost is constant, so MC is a flat line. The high fixed cost is spread over more cauldrons as output grows, so $ATC = MC + FC/q$ starts high and falls toward MC without ever reaching it.

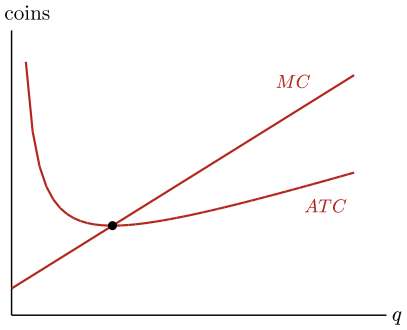
The Witching Hour. Marginal cost rises with output, so MC slopes up. At low output the fixed cost dominates and ATC falls; once MC rises above ATC it pulls the average up. ATC is U-shaped and MC crosses it at its lowest point.

Nimbus Cauldron Works. With no fixed cost and a constant marginal cost, every cauldron costs the same to make, so the average equals the marginal: $ATC = MC$, one flat line.

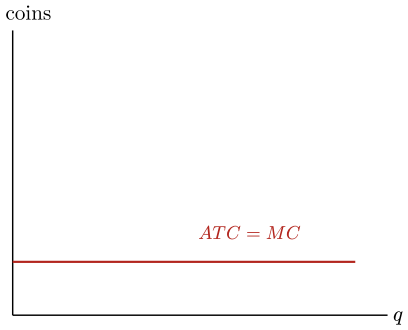
The Cauldron Co.



The Witching Hour



Nimbus Cauldron Works

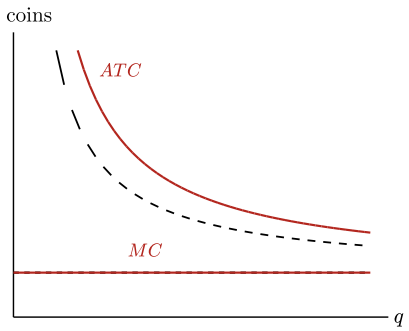


Q2 | Lump Sum Tax

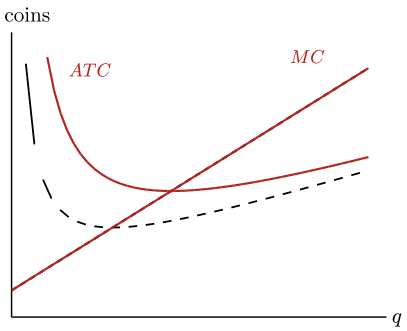
Model a lump sum tax on each of the three firms.

> A lump sum tax is the same amount however much the firm makes, so it acts like a fixed cost. Marginal cost does not move. Average total cost rises by the tax divided by output, so it rises most at low output and the gap closes as output grows. Nimbus had no fixed cost; now its ATC sits above its marginal cost and falls toward it. The Witching Hour's ATC still meets MC at its lowest point, which moves to a larger output. (Drawn with a tax of 300 coins.)

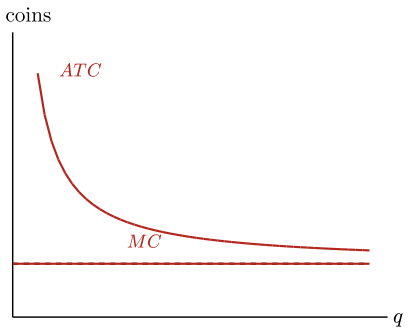
The Cauldron Co.



The Witching Hour



Nimbus Cauldron Works

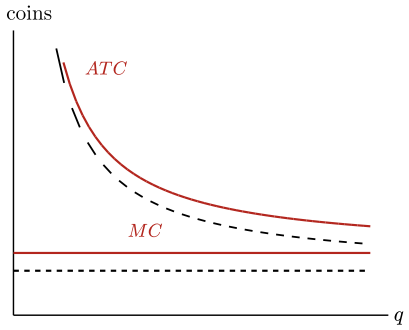


Q3 | Per Unit Tax

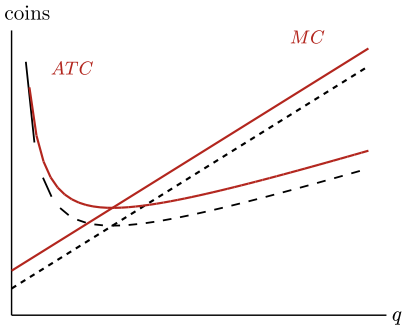
Model a per unit (*called specific*) tax on each of the three firms.

A per unit tax is paid on every cauldron, so every cauldron costs that much more to make. Marginal cost and average total cost both shift up by the tax, at every output. The shapes do not change, and The Witching Hour's lowest ATC stays at the same output. (Drawn with a tax of 4 coins per cauldron.)

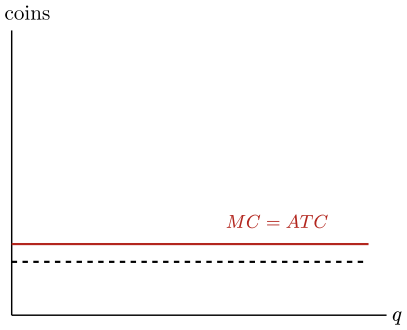
The Cauldron Co.



The Witching Hour



Nimbus Cauldron Works

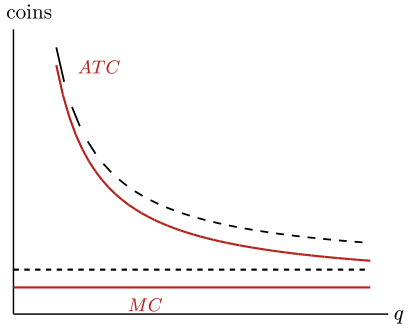


Q4 | Per Unit Subsidy

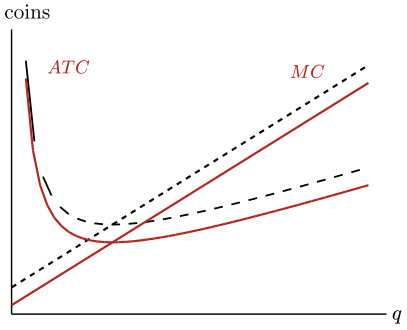
Model a per unit (*called specific*) subsidy on each of the three firms.

A per unit subsidy pays the firm on every cauldron, so every cauldron costs that much less to make. Marginal cost and average total cost both shift down by the subsidy, at every output. (Drawn with a subsidy of 4 coins per cauldron.)

The Cauldron Co.



The Witching Hour



Nimbus Cauldron Works

